nature portfolio

https://doi.org/10.1038/s41586-022-05352-2

Supplementary information

Singlet and triplet Cooper pair splitting in hybrid superconducting nanowires

In the format provided by the authors and unedited

Supplementary Information

Contents

1 SOC in QDs **1**

2 g-factor anisotropy **3**

3 Theoretical modelling **4**

3.1 CAR and ECT resonant currents 4

3.2 QD-QD coupling strengths 5

3.3 Parameters in the model 7

3.4 Analytical formula of angular oscillation amplitudes 8

4 References **8**

1 SOC in QDs

We have interpreted the measured anisotropy in CAR/ECT amplitudes in the Main Text as resulting from SOC in the hybrid section. It warrants discussion that the QDs themselves do not have purely spin up or down states in the presence of B, either, because of SOC [1]. Fig. ED3 shows that along y, the spin-orbit field direction of the QD, spin is a good quantum number and excited states belonging to different orbitals and opposite spins cross each other. When $B = B_x$ is perpendicular to the QD spin-orbit field, QD eigenstates become admixtures of $|n, \uparrow\rangle$ and $|n+1, \downarrow\rangle$ (n being the orbital index number of the

QD) and nominally opposite spin states anti-cross. Qualitatively, this produces the same effect that spin blockade is complete when $B \parallel B_{\mathrm{SO}}$ in both QDs and gets lifted in other directions because of the mixing even if the hybrid segment is purely spin-singlet. However, the estimation below of the strength of this effect shows it plays a small role in our observed oscillations.

The amount of spin-up mixed into the nominally spin-down QD level is $\langle H_{\mathrm{SO}}\rangle/(\delta - E_{\mathrm{Z}})$ according to first-order perturbation theory [1], where $2\langle H_{\mathrm{SO}}\rangle$ is the spin-orbit level repulsion gap between the two mixing orbitals, δ the orbital level spacing between them and E_{Z} the Zeeman splitting. All energies can be directly read out from bias spectroscopy of the QD excitation energies as a function of B (Fig. ED3b,d). The largest value of $\langle H_{\mathrm{SO}}\rangle/(\delta - E_{\mathrm{Z}})$ we measured is seen in Fig. ED3b, where the spin-orbit level repulsion is 0.4 meV and the difference between Zeeman splitting and level spacing is 1.6 meV. We thus observe no more than $0.2/1.6 = 12.5\%$ of the opposite admixture at $B = 100\,\mathrm{mT}$, namely the nominally $|\uparrow\rangle$ state becomes modified as $|\uparrow\rangle + 0.125\,|\downarrow\rangle$ via perturbation. If for each QD, the nominally spin-up level has a spin composition of $\alpha_\eta\,|\uparrow\rangle + \beta_\eta\,|\downarrow\rangle$ and the nominally spin-down is $\beta_\eta\,|\uparrow\rangle - \alpha_\eta\,|\downarrow\rangle$ where $\eta = \mathrm{L}, \mathrm{R}$ and coefficients are normalized, the spin-flipping probability produced by SOC in QD is $|\alpha_{\mathrm{L}}\beta_{\mathrm{R}} + \alpha_{\mathrm{R}}\beta_{\mathrm{L}}|^2$. Using the worst-case-scenario numbers extracted from Fig. ED3b above for both QDs, the expected triplet current at $\phi = 0°, 180°$ should be no more than 6% of the corresponding singlet value if the superconducting hybrid has no SOC. Given that our observed triplet current at $\phi = 0°, 180°$ is between 20% and 50% of the spin-singlet counterpart, the majority of the contribution must originate from SOC in the hybrid. This is corroborated by the fact that a moderate value of SOC taken from the literature for the Al-InSb segment reproduces the measured features and that the maximum spin-blockade direction is roughly 18° out of plane for

Device A and 30° for Device B, consistent with the three-facet Al coverage on InSb. We will also present data on the gate tunability of B_{SO} in a future manuscript to substantiate this interpretation[1].

The conclusion that SOC effects are much stronger in the hybrid—despite renormalization of InSb parameters by hybridization with Al [2]—has several possible explanations. Firstly, the length of the hybrid segment is more than twice the QD size, leading to more spin precession as electrons traverse through it. In addition, band bending at the Al-InSb interface may lead to much stronger electric fields [2] than produced by gate electrodes in the QDs.

Although this confirmation of the relative SOC strengths is helpful to asserting the p-wave nature of the hybrid superconducting pairing, triplet Cooper pair splitting can indeed also be achieved via two strongly spin-orbit-coupled QDs connected by a purely s-wave superconductor under B, e.g. in the situation where the QD dimensions are larger than the hybrid segment. The key to inducing triplet correlations on two QDs is to introduce non-collinear spin quantization axes in QDs and in the hybrid. Thus in the semiclassical description, injecting a spin pointing along a different direction than the Zeeman field leads to precession. The two scenarios are, in fact, physically equivalent in this aspect considering that SOC during tunneling and spatially varying Zeeman quantization axes only differ by a gauge transformation [3].

2 g-factor anisotropy

InSb nanowire QDs have anisotropic g factors [4]. For essentially the same reason as presented in [4], g anisotropy is not a plausible cause of the observed spin blockade and lifting behavior. In summary, an anisotropic g tensor has three principal axes and rotating B in one plane generally encounters two

[1] Bordin, et al., in preparation

of them. Along both axes, the spins in the dots are aligned and blockade is complete. This results in two peaks and dips in 180° of rotation instead of one, inconsistent with our observed periodicity.

3 Theoretical modelling

We present here the theoretical model of the experimental data presented in this work. Details of the model can be found in the parallel theory work [5].

3.1 CAR and ECT resonant currents

In the Main Text, the resonant current flowing through the N-QD-S-QD-N hybrid system is measured. Such CAR and ECT resonant currents have the forms [5]

$$\begin{aligned} I_{\mathrm{CAR}} &= \frac{e}{\hbar} \cdot \frac{\Gamma_{DL}^2}{(\varepsilon_l + \varepsilon_r)^2 + \Gamma_{DL}^2} \cdot \frac{\Gamma_{\mathrm{CAR}}^2}{\Gamma_{DL}}, \\ I_{\mathrm{ECT}} &= \frac{e}{\hbar} \cdot \frac{\Gamma_{DL}^2}{(\varepsilon_l - \varepsilon_r)^2 + \Gamma_{DL}^2} \cdot \frac{\Gamma_{\mathrm{ECT}}^2}{\Gamma_{DL}}, \end{aligned} \tag{1}$$

respectively. Here Γ_{DL} is the total QD-N coupling strength, i.e. sum of left-QD–left-N coupling and right-QD–right-N coupling. Γ_{CAR} and Γ_{ECT} are the effective CAR and ECT coupling between the two QDs. These couplings depend on the properties of the middle hybrid segment and on the spin polarization in the QDs, but do not depend on the energies of the QDs. Thus in the $(\varepsilon_l, \varepsilon_r)$-plane, the resonant current assumes a Lorentzian shape with the broadening being Γ_{DL} and reaching the maximum value along $\varepsilon_l = \pm\varepsilon_r$ for ECT and CAR, respectively. Thereby the maximum current for CAR and ECT are

$$I_a^{\max} = \frac{e\Gamma_a^2}{\hbar\Gamma_{DL}} \propto \Gamma_a^2 \propto P_a, \tag{2}$$

where a = CAR or ECT [5]. Here $P_{\mathrm{CAR/ECT}}$ are the transition probability of electrons between two dots: P_{CAR} is the probability of two electrons of a Cooper pair tunneling from the middle superconductor to the two separate dots and P_{ECT} is the probability of a single electron tunneling from the left to the right QD.

Fitting the correlated currents in Fig. 2 of the Main Text to Equation (1), we obtain total QD-N tunnel couplings of 57 (panel d) and 29 µeV (panel e) and QD-QD couplings of 9.6 (panel d) and 3.8 µeV (panel e).

3.2 QD-QD coupling strengths

The Bogoliubov-de-Gennes (BdG) Hamiltonian for the one-dimensional semiconductor-superconductor hybrid nanowire is

$$\begin{aligned} H_0 =& \frac{1}{2}\int_0^L dx \Psi^\dagger(x) h_0(x) \Psi(x), \\ h_0(x) =& \left(-\frac{\hbar^2}{2m^*}\partial_x - \mu\right)\tau_z + i\alpha_R \partial_x \left(\cos\theta \tau_z \sigma_y + \sin\theta \sigma_z\right) \\ &+ E_Z \tau_z \sigma_z + \Delta_{\mathrm{ind}} \tau_y \sigma_y, \end{aligned} \tag{3}$$

where $\Psi(x) = [c_\uparrow(x), c_\downarrow(x), c_\uparrow^\dagger(x), c_\downarrow^\dagger(x)]^\intercal$ is the Nambu spinor, τ_α and σ_α are Pauli matrices acting on the Nambu and spin space, L is the length of the hybrid nanowire, m^* is the effective mass of InSb, μ is the chemical potential, α_R is the strength of Rashba spin-orbit coupling, E_Z is half Zeeman spin splitting, θ is the angle between the spin-orbit field and the Zeeman field, and Δ_{ind} is the induced superconducting pairing strength in the nanowire. In the limit of confinement in all three dimensions as is the case with Device A, Δ_{ind} can be read as the lowest energy that the discrete superconducting-semiconducting hybrid states reach. We define the direction of the Zeeman field to be always along σ_z, thus it is the spin-orbit field that rotates in our reference frame.

The tunneling Hamiltonian between the hybrid nanowire and the QDs is

$$H_T = \left(-t_l c_\eta^\dagger(0) d_{l,\eta} - t_r c_\sigma^\dagger(L) d_{r,\sigma}\right) \tau_z + \text{H.c.}, \tag{4}$$

where $d_{r,\sigma}$ and $d_{l,\eta}$ are the annihilation operators of the spin-polarized state in QD-l and QD-r. Here we assume that the single-electron tunneling is spin-conserving, and that there is a single QD level with a particular spin polarization (along σ_z) near the Fermi energy of the normal QDs. That is, the dot state in the left QD has spin-σ and that in the right QD has spin-η. In the tunneling limit, i.e., $t_{l,r} \ll \Delta_{\text{ind}}$, the effective CAR or ECT tunneling between the two dots is well described by the second-order virtual process $\langle H_T H_0^{-1} H_T \rangle$ where $\langle \cdot \rangle$ denotes the ground state of the hybrid nanowire. For example, the effective CAR coupling between spin-η state in QD-l and spin-σ state in QD-r is

$$\begin{aligned}
\Gamma_{\text{CAR},\eta\sigma} &= t_l t_r \left(\langle c_\eta^\dagger(0) H_0^{-1} c_\sigma^\dagger(L)\rangle - \langle c_\sigma^\dagger(L) H_0^{-1} c_\eta^\dagger(0)\rangle\right) \\
&= \frac{t_l t_r}{\Delta_{\text{ind}}} \sum_m \frac{u_{m\eta}(0) v_{m\sigma}^*(L) - u_{m\sigma}(L) v_{m\eta}^*(0)}{E_m/\Delta_{\text{ind}}} \\
&= \frac{t_l t_r}{\Delta_{\text{ind}}} a_{\eta\sigma},
\end{aligned} \tag{5}$$

where $[u_{m\uparrow}(x), u_{m\downarrow}(x), v_{m\uparrow}(x), v_{m\downarrow}(x)]^\intercal$ is the Bogoliubov wavefunction of the m-th eigenstate with excitation energy $E_m > 0$. Thereby for the resonant current flowing through the normal QDs with particular spin polarizations, we have

$$I_{\text{CAR},\eta\sigma}^{\max} \propto \left(\Gamma_{\text{CAR},\eta\sigma}\right)^2 \propto P_{\text{CAR},\eta\sigma} = |a_{\eta\sigma}|^2. \tag{6}$$

Similarly, for the ECT current, we have

$$\begin{aligned}\Gamma_{\mathrm{ECT},\eta\sigma} &= t_l t_r \left(\langle c_\eta(0) H_0^{-1} c_\sigma^\dagger(L)\rangle - \langle c_\sigma^\dagger(L) H_0^{-1} c_\eta(0)\rangle\right) \\ &= \frac{t_l t_r}{\Delta_{\mathrm{ind}}} \sum_m \frac{u_{m\eta}(0) u_{m\sigma}^*(L) - v_{m\sigma}(L) v_{m\eta}^*(0)}{E_m/\Delta_{\mathrm{ind}}} \\ &= \frac{t_l t_r}{\Delta_{\mathrm{ind}}} b_{\eta\sigma} \end{aligned} \tag{7}$$

and

$$I_{\mathrm{ECT},\eta\sigma}^{\max} \propto \left(\Gamma_{\mathrm{ECT},\eta\sigma}\right)^2 \propto P_{\mathrm{ECT},\eta\sigma} = |b_{\eta\sigma}|^2. \tag{8}$$

For the numerical simulation, we first discretize the Hamiltonian in Eq. (3) into a tight-binding Hamiltonian using KWANT [6]. We then get the eigenenergies and eigenfunctions by diagonalizing the Hamiltonian and calculate the probabilities using Eqs. (5) and (7).

3.3 Parameters in the model

The physical parameters we choose for the BdG Hamiltonian in Eq. (3) are: $L = 200\,\mathrm{nm}$ for Device A and $L = 350\,\mathrm{nm}$ for Device B, $m^* = 0.015 m_e$ for InSb, $\alpha_R = 0.15\,\mathrm{eV\AA}$ as estimated using oscillation amplitudes. The bare superconducting gap in the parent superconductor is $\Delta_0 = 0.3\,\mathrm{meV}$, extracted from tunnel spectroscopy in Fig. ED9. The renormalized g-factor of the hybrid state is estimated to be around 20 from Fig. ED9. This value being half that of InSb also implies the induced superconducting pairing strength Δ_{ind} is around half the parent gap [2], around $0.15\,\mathrm{meV}$. At $B = 0.1\,\mathrm{T}$, the estimated Zeeman field strength is $E_Z = \frac{1}{2} g \mu_B B \approx 0.06\,\mathrm{meV}$. After discretizing the continuum Hamiltonian into the lattice model, a chemical potential fluctuation with

amplitude $\delta\mu = 10\,\mathrm{meV}$ is added on each site to model a moderate amount of disorder in the nanowire.

3.4 Analytical formula of angular oscillation amplitudes

The main use of the numerical simulations is to compare with the measured up-up/up-down CAR amplitudes to obtain an estimation of the SOC strength α_R. Ref [5] also contains derivations of an analytical expression that directly relates α_R to the angle-averaged up-up/up-down ratio of CAR amplitudes:

$$\frac{\overline{P}_{\mathrm{CAR},\uparrow\uparrow}}{\overline{P}_{\mathrm{CAR},\uparrow\downarrow}} = \frac{\sin^2(k_{so}L)}{2-\sin^2(k_{so}L)} \tag{9}$$

where $k_{so} = m\alpha_R/\hbar^2$ and L is device length. This expression is insensitive to microscopic details of the wavefunction in the hybrid produced by varying chemical potential, disorder and inhomogeneous α_R. It depends only on the nanowire length and the averaged spin-orbit length. The ratio is no greater than one and reaches the maximum when the nanowire length is a half-integer multiple of the spin-orbit length. As we can see in Fig. ED5, the difference between numerical simulations and Eq. (9) is indeed small. The deviations mostly originate from finite Zeeman field not captured by the analytical formula. Thus, this simple expression provides us with a way of extracting the spin-orbit constant α_R directly from the anisotropic CAR/ECT measurements without any other knowledge than device length and effective electron mass.

4 References

[1] Hanson, R., Kouwenhoven, L.P., Petta, J.R., Tarucha, S., Vandersypen, L.M.K.: Spins in few-electron quantum dots. Rev. Mod. Phys. **79**, 1217–1265 (2007). https://doi.org/10.1103/RevModPhys.79.1217

[2] Antipov, A.E., Bargerbos, A., Winkler, G.W., Bauer, B., Rossi, E., Lutchyn, R.M.: Effects of gate-induced electric fields on semiconductor majorana nanowires. Physical Review X **8**(3), 031041 (2018)

[3] Braunecker, B., Japaridze, G.I., Klinovaja, J., Loss, D.: Spin-selective Peierls transition in interacting one-dimensional conductors with spin-orbit interaction. Physical Review B **82**(4), 045127 (2010)

[4] Nadj-Perge, S., Pribiag, V.S., van den Berg, J.W.G., Zuo, K., Plissard, S.R., Bakkers, E.P.A.M., Frolov, S.M., Kouwenhoven, L.P.: Spectroscopy of spin-orbit quantum bits in indium antimonide nanowires. Phys. Rev. Lett. **108**, 166801 (2012). https://doi.org/10.1103/PhysRevLett.108.166801

[5] Liu, C.-X., Wang, G., Dvir, T., Wimmer, M.: Tunable superconducting coupling of quantum dots via andreev bound states. arXiv preprint arXiv:2203.00107 (2022) 2203.00107

[6] Groth, C.W., Wimmer, M., Akhmerov, A.R., Waintal, X.: Kwant: a software package for quantum transport. New Journal of Physics **16**(6), 063065 (2014)